MANIPAL UNIVERSITY

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School of Computer Science and Engineering

Department of Computer Science and Engineering

AI-Based Financial Expense Tracker: FinAi

Submitted By:
Name : Pratham Mehriya
Registration no. : 2427030696

Supervised By:
Name: Tripti Kulshrestha

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Introduction

AI-Based Financial Expense Tracker

- Tracks all your expenses automatically and categorizes them for clarity.
- Visualizes spending patterns through interactive charts and graphs.
- Uses AI to predict potential overspending and highlight areas to save.
- Provides personalized tips to manage money more efficiently.
- Helps users make informed financial decisions and build healthier money habits.
- Turns budgeting from a stressful task into a simple, smart, and stress-free experience.

Literature Review

1. Traditional expense trackers rely mainly on manual entry and simple categorization.
2. With the advent of AI, predictive analytics can now identify overspending patterns.
3. Studies show that visualization and automated insights improve user engagement.
4. Existing tools lack personalized recommendations based on spending history.

Feature	Walnut	PocketGuard	FinAI
Expense Tracking	✓	✓	✓
Income Management	✓	✓	✓
Visual Analytics (Charts)	✓	✓	✓
Budget Limit Alerts	✓	✓	✓
Conscious Consumption	✗	✗	✓
AI Moral Insight Engine	✗	✗	✓
Ethical Spending Coaching	✗	✗	✓
“Wealth “Karma” Tips”	✗	✗	✓
Behavioral Ratio Analysis	✗	✗	✓
Automated Bank Syncing	✓	✓	✗
Multi-Currency Support	✓	✓	✗

Comparing FinAi with other Finance Trackers.

Problem Statement

- Most expense tracking systems lack intelligent analysis and predictive insights, making it difficult for users to identify overspending patterns and plan savings effectively.
- Most of them only record expenses but do not help users analyse habits, plan savings, or make responsible financial decisions.
- Current systems do not encourage responsible habits such as saving regularly, or contributing to social causes.

Proposed Solution

1. Collect and categorize user expenses
2. Visualize spending patterns for monthly insights.
3. Employ predictive models to forecast potential overspending.
4. Provide AI-driven suggestions such as: 'Reduce spending on food to save ₹2,500.'
5. Provide insights on expensed

Methodology

The development of FinAI follows a structured approach to ensure efficient data handling, analysis, and user interaction. The methodology consists of the following steps:

- **Data Collection**: User inputs financial data such as income and expenses through the interface.
- **Data Categorization**: Transactions are classified into categories like food, rent, and entertainment for better organization.
- **Data Processing**: The system calculates total income, expenses, and balance using JavaScript logic.
- **Data Visualization**: Charts and graphs are generated using Chart.js to represent spending patterns.
- **Analysis & Insights**: The system analyzes user behavior to identify overspending trends.
- **Recommendation Generation**: AI-based suggestions are provided to help users improve financial habits.

Objectives

- To develop an intelligent financial dashboard using AI and Python.
- To analyze and visualize monthly spending patterns.
- To predict future expenses using machine learning models.
- To generate actionable insights that help users save money.

FinAI

📊 Dashboard

🎯 Budget Goals

🤖 AI Advisor

Dashboard

Currency: Indian Rupee (₹)

+ New Entr

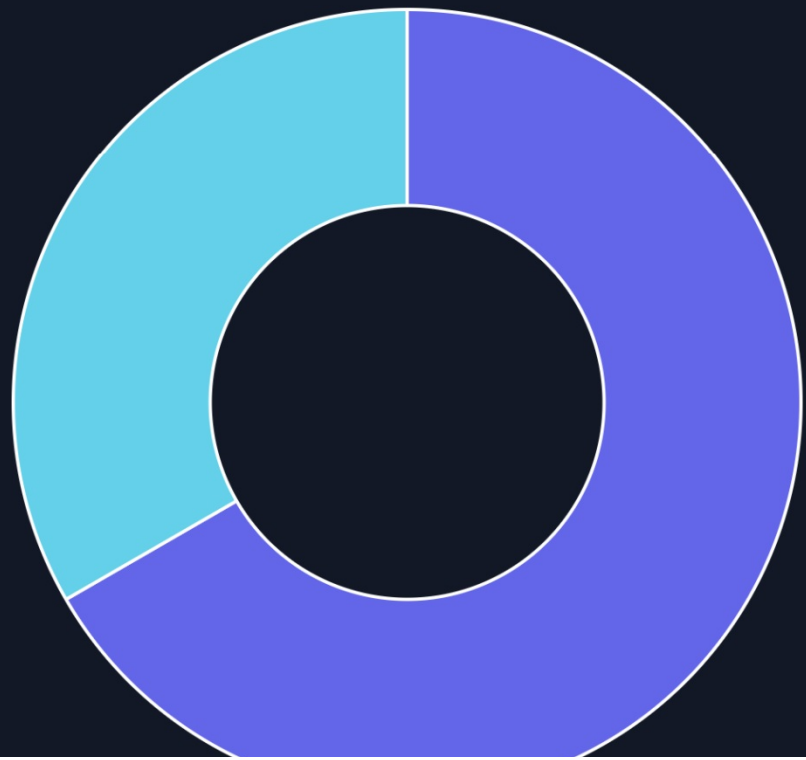
CURRENT BALANCE
₹94,000

TOTAL INCOME
₹1,00,000

TOTAL EXPENSES
₹6,000

Monthly Trends

Category Comparison



Dashboard

Budget Goals

AI Advisor

Budget

Currency: Indian Rupee (₹)

+ New Entry

Set Category Limits

Food

5000

Set Goal

Entertainment ₹0 / ₹2000

Shopping ₹4000 / ₹3000

Food ₹2000 / ₹5000

References

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Thank You